

SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID – 26125

Problem Statement Title - Blockchain-Based Secure Platform for Identity, Access Control, and Digital Asset Management

Theme - Blockchain & Cybersecurity

PS Category - Software

Team ID - [Team ID]

Team Name - Cyber Samurai's



IDEA TITLE

❖ Decentralized Identity, Access & Digital-Asset Platform on a Redactable Blockchain

💡 Proposed Solution

- Unifies **decentralized identity (DID)**, **role-based access** and **NFT-based asset ownership** into one trustless platform.
- Every identity, permission, mint & transfer is **immutably recorded on-chain**.
- Hardened with **RFID physical-presence auth + AES-256** encryption.

🎯 How it addresses the problem

- Removes the **single point of failure** of centralized IAM.
- Cryptographic IDs stop **identity theft & unauthorized access**.
- Smart contracts govern **who can mint / allocate / transfer** assets.
- On-chain trail makes ownership & access **verifiable & tamper-evident**.

★ Innovation & Uniqueness

- **Real chameleon-hash** redaction — lawful correction without breaking the chain (genuine crypto, not a simulation).
- **Hardware step-up auth** — ESP32 + RFID for privileged actions.
- **Consent-augmented RBAC** — role + purpose-limited access.

TECHNICAL APPROACH

React + TypeScript

Flask · Python 3.13

MongoDB (AES-256)

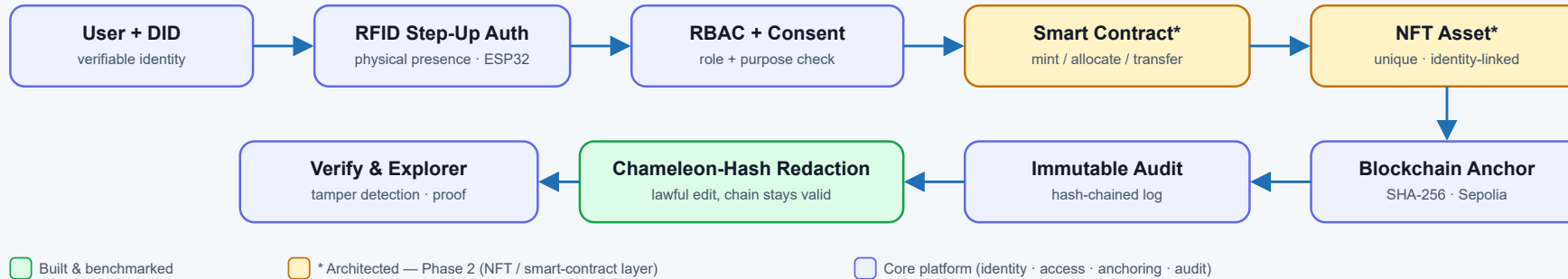
Web3.py · Ethereum Sepolia

Solidity / ERC-721*

JWT + bcrypt

ESP32 + RC522 RFID

Docker · CI



FEASIBILITY AND VIABILITY

Feasibility of the idea

- **Real** 1536-bit discrete-log chameleon hash (RFC 3526)
- **164 automated tests** pass on every push (CI)
- Deployable: Render + Vercel + MongoDB Atlas
- Public verifiability via Ethereum **Sepolia**

Economic & operational viability

- 100% **open-source** stack — no licence cost
- **₹0 infrastructure** on managed free tiers
- Testnet anchoring = **no gas cost** for demo
- Containerized (Docker) — portable to any cloud

Challenges & mitigation

- Trapdoor key safety → **HSM / KMS** (roadmap)
- On-chain PII risk → **hashes only**, never raw data
- Single-node trust → **public Sepolia** chain
- NFT/DID layer → phased, standards-based (ERC-721, W3C DID)

Measured performance — Chameleon Hash (1536-bit, mean of 100 runs)

~0.25 ms

Authorized redaction — ~4,000 ops/sec, constant vs record size

~13 ms

Hash / Verify — size-independent (SHA-256 pre-compression)

~9 ms

Key generation — one-time trapdoor setup

50× faster

Redaction vs hashing — lawful edits practical at scale

IMPACT AND BENEFITS

Potential impact on target audience

- **Eliminates single point of failure** — no central identity DB to breach.
- **Tamper-proof ownership** — asset & access history is cryptographically verifiable.
- **Privacy by design** — AES-256 + data-protection rights.
- **Trust without intermediaries** — smart contracts enforce rules automatically.

Beneficiaries

- **Enterprises** — secure workforce identity & access governance.
- **Government** — asset registries, licences, credential issuance.
- **Regulated sectors** — auditable, consent-driven records.
- **Individuals** — self-sovereign identity & provable ownership.

Social benefit

- User control & data-protection rights
- Transparent, trustable institutions
- Inclusive self-sovereign identity

Economic benefit

- Cuts fraud & identity-theft losses
- Removes costly central IAM & reconciliation
- Open-source, low deployment cost

Security benefit

- Decentralized cryptographic identity
- Hardware-backed step-up auth
- Immutable, verifiable audit trail

RESEARCH AND REFERENCES

References & standards

- Krawczyk & Rabin — *Chameleon Hashing & Signatures*, NDSS 2000
- Ateniese et al. — *Redactable Blockchain*, IEEE EuroS&P 2017
- W3C — *Decentralized Identifiers (DID) v1.0*
- Ethereum — *ERC-721 Non-Fungible Token Standard*
- RFC 3526 — MODP Diffie-Hellman groups (1536-bit)
- Digital Personal Data Protection Act, 2023 (India)

Evidence & project links

- 164 automated tests · CI pipeline green
 - Reproducible cryptographic benchmark harness
 - Working ESP32 + RC522 RFID hardware terminal
 - Documented threat model & compliance mapping
-  **GitHub:** github.com/Bennyhinn007/...Consent-Management-System
-  **Live demo:** [\[deployment link\]](#)